INCH-POUND

MIL-C-24643/34C
14 March 1994
SUPERSEDING
MIL-C-24643/34B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS TYPE LSMS (INCLUDING VARIATION LSMSA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSMS (unarmored) and LSMSA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG no. 16, class B stranding, (7 by 0.0192 inch) or in accordance with ASTM B 186, size 16-19, tin coated. Maximum diameter 0.063 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Ethylene propylene rubber or cross-linked polyethylene insulation, white. Minimum average wall thickness 0.013 inch. Standard identification code applied by method 1.

Fourth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter. Cabling sequence shall be consecutive starting with no. 1 from center outward.

Fifth - Binder tape, applied helically with overlap.

* Sixth - Braided shield of tin coated copper. Minimum coverage of 85 percent.

Construction variant for type LSMS

* Seventh - Binder tape applied helically with overlap.

* Eighth - Cross-linked polyolefin jacket. Minimum average cable jacket thickness 0.075 inch.

AMSC N/A FSC 6145

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Construction (non-watertight) (continued)

Construction variant for type LSMA

* Seventh - Binder tape applied helically with overlap.
* Eighth - Cross-linked polyolefin jacket. Minimum average cable jacket thickness 0.075 inch.
Ninth - Braided metal armor.

TABLE I. Details.

Military 1/ part no. M24643/34	Size 2/	Number of Conductors	Diameter over Shielded core (inch nom.)	Overall diameter (inches)			
				TYPE LSMS		TYPE LSMSA	
				min	max	min	max
-01UO	37	37	0.625	0.740	0.800	0.790	0.850

1/ Unarmored, overall shield. When armor option or double shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LSMS AND LSMSA.

NOTE: Weight of completed cable shall not exceed 0.060 pound/Foot.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum ...	5.1
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulator resistance - Megohms/1000 feet minimum	
Conductor to conductor	2000
Conductor to shield	500
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470) ...	35

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EXAMINATION AND TESTS (continued):	Requirements
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
*Permanence of printing (conductor) - cycles minimum	25
*Permanence of printing (jacket - (types LSMS only) cycles, minimum	125
Armor (type LSMSA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Surface transfer impedance (type LSMUS only)	
Milliohms per meter, maximum	700
Emp response time - db, minimum	60
*Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
* Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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